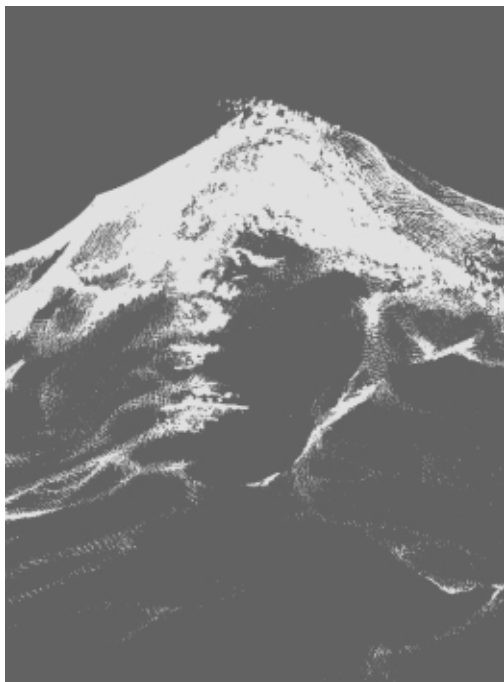


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Size and reproductive status of fished spotted spiny lobster, *Panulirus guttatus*, in Morrocoy National Park, Venezuela: a preliminary report

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Abstract. The Venezuelan lobster-management plan is based on information collected from the *Panulirus argus* fishery in Los Roques Archipelago National Park. *Panulirus guttatus* are harvested illegally, as this species does not reach the minimum landing size (120 mm carapace length; CL). Fishers consider enforcement of a 120 mm minimum CL unjust and request a new regulation. The purpose of this study was to determine the basic population characteristics of *P. guttatus* at the Morrocoy National Park, to provide information for managers, and to establish a population monitoring program. Interviews with fishers were conducted from September to December 1999, and their catch was counted, measured, and sexed. Of the 163 lobsters surveyed, 93.3% were *P. guttatus*, 6.1% *P. argus*, and 0.6% *P. laeviscauda*. The size range for *P. guttatus* was 42.1 to 75.6 mm CL (average: 60.9 ± 6.9 mm CL). Sex ratio favoured males (1.45:1), and 48.4% of the 62 females examined bore eggs. The smallest reproductive female measured 42.1 mm CL; the largest measured 67.7 mm CL. Fishing is carried out in shallow waters of the outer reefs; lobsters are harvested by hand during nocturnal skin diving.

Extra keywords: management plan, Palinuridae, Caribbean, spiny lobster fishery, fishery legislation

Introduction

Spiny lobsters of the family Palinuridae are abundant in nearly all the tropical and subtropical oceans of the planet. They are found in the western Atlantic, from North Carolina (USA) and Bermuda to Brazil, including the Gulf of Mexico and the Caribbean Sea (Padilla-Ramos and Briones-Fourzán 1997; Tewfik *et al.* 1998). In the Caribbean, this family is best represented by three species: *Panulirus argus* (Latreille, 1804), *P. guttatus* (Latreille, 1804), and *P. laeviscauda* (Latreille, 1817). *P. argus* has been most widely studied, because of its higher abundance, larger size, and commercial importance. Most of the legislation formulated to control lobster fishing is therefore based on knowledge of the life history and fishery for *P. argus* (Briones-Fourzán 1995).

In Venezuela, the lobster fishery is exclusively a small-scale artisanal activity, developed mainly on the insular platform of the Los Roques and Los Testigos archipelagos and on the continental shelf of Venezuela's western coast (Hauschild and Laughlin 1985). The *P. argus* fishery in Los Roques Archipelago National Park has been the most studied (Cobo de Barany *et al.* 1972; Ginéz *et al.* 1978; Hauschild and Weil 1983; Hauschild and Laughlin 1985; Posada *et al.* 1996); it is the basis for a national management plan, which includes a seasonal closure (from 1 May to 31

October) and a minimum landing size of 120 mm carapace length (CL).

Traditionally, Morrocoy National Park authorities confiscated all lobsters whose maximum length was below current regulations. The Morrocoy fishing community indicated that individuals captured below the minimum allowable length belong to a species that never reaches this minimum landing size, and they therefore consider this legal practice unjust. Subsequent investigations revealed that their catch was composed of individuals of the spotted spiny lobster, *Panulirus guttatus*.

The biology, ecology, and economic importance of *P. guttatus* differ in several key ways from those of *P. argus* (Sutcliffe 1953; Colinas-Sánchez and Briones-Fourzán 1990; Briones-Fourzán 1995; Evans *et al.* 1995b; Sharp *et al.* 1997; Briones-Fourzán and Contreras-Ortiz 1999), aside from just smaller size (maximum of 88 mm CL; Briones-Fourzán 1991). For example, the commercial importance of *P. guttatus* is limited, except in Bermuda, Martinique, and the French West Indies (Evans and Lockwood 1994). This observation may be biased, however, as landing records in most of the Caribbean countries do not differentiate between species of *Panulirus*. Nevertheless, these differences are starting to be acknowledged, and it is known that 5 to 10% of the lobsters captured in Mexico (Briones-Fourzán 1995),

and 43% of the lobsters caught in Bermuda (Evans *et al.* 1995a), are *P. guttatus*.

The present study provides a preliminary perspective on population characteristics and a description of a small artisanal fishery for *P. guttatus* at the Morrocoy National Park, Venezuela. The results indicate the need for both new fishing regulations and a population-monitoring program for *P. guttatus*.

Methods

The study area was located in the Morrocoy National Park, on the western coast of Venezuela. The park extends from the east of Tucacas to Chichiriviche, between latitudes 10°47'12" and 10°55'30"N and longitudes 68°14'0" and 68°14'32"W (Fig. 1). The National Institute of Parks (INPARQUES) requested the initial evaluation of the conflictive

fishery for *P. guttatus* that is taking place in the Morrocoy National Park. Preliminary surveys were carried out from September to December 1999 and consisted of direct interviews with fishers at their houses or landing sites. Most of the fishers for *P. guttatus* were located in the town of Tucacas. The number of active fishers in this community was ~23 (Klein and Salaya 1999), 10 to 14 of which fish for *P. guttatus*. Most of these 23 fishers augment their income working temporarily in tourism and construction activities. Of the estimated 10 to 14 fishermen that concentrate on *P. guttatus*, eight were interviewed.

Fishers were interviewed only once. The interviews were designed to evaluate rapidly the basic features of the fishery. During each interview we also sampled the catch to quantify lobster population characteristics. During interviews with fishers the following data were collected: most frequently visited fishing areas; durations and times of fishing, and depth and method of capture. Population characteristics included length (carapace, abdominal and total, all to the nearest

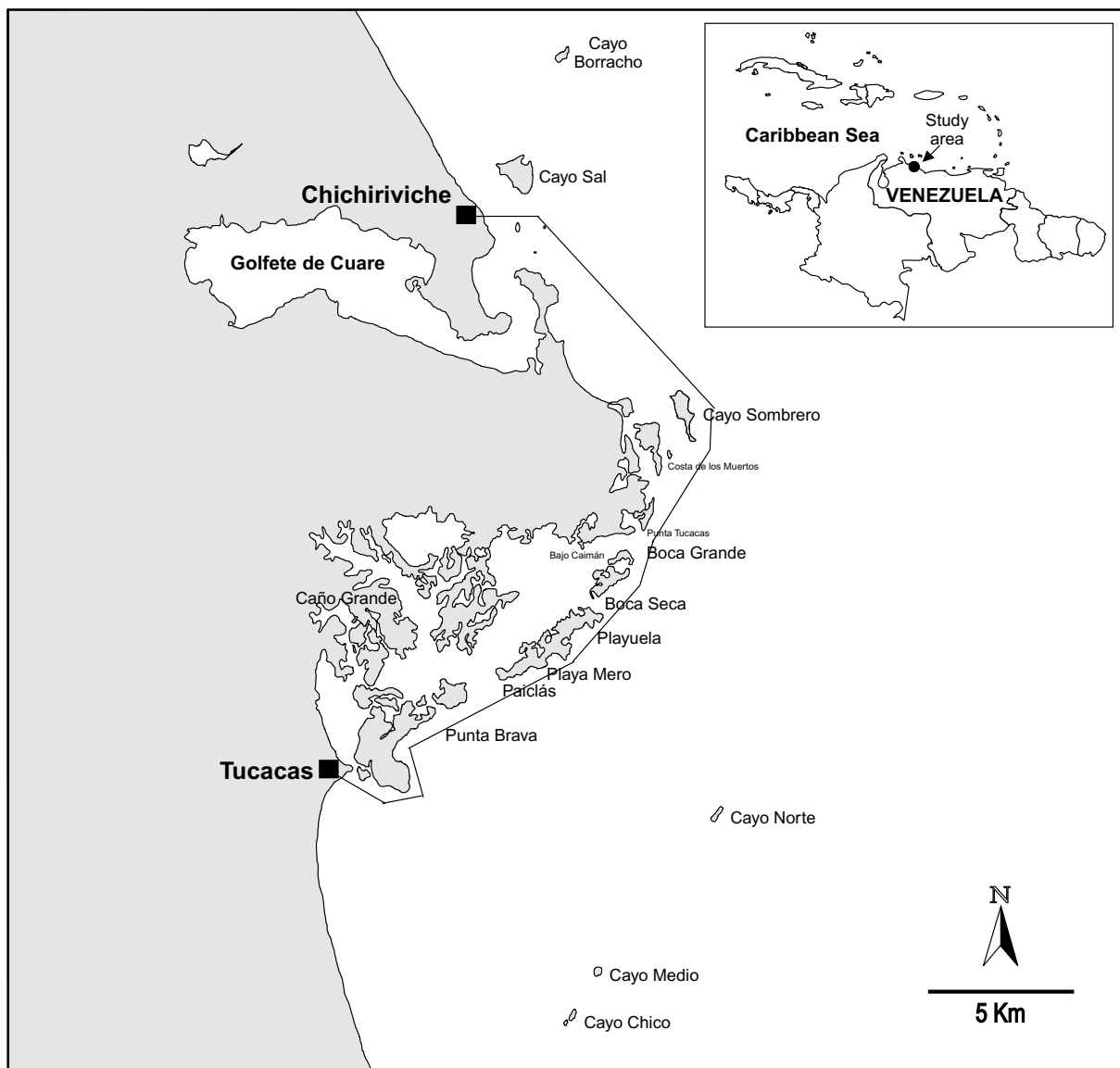


Fig. 1. Map of the Morrocoy National Park, Venezuela, showing fishing areas for *P. guttatus* between the towns of Tucacas and Chichiriviche.

millimetre), sex, and reproductive condition of females (e.g., egg-bearing).

Results

Fishery Characteristics

Lobsters are captured by hand during nocturnal skin diving, especially during the first lunar quarter. Fishers usually go out in groups of 4 to 5 individuals, leaving just before sunset (about 1700 to 1800 hours) and wait in the fishing area until dark. Fishing areas are located in the windward reefs of the outer keys (Fig. 1), at depths no greater than 5 m. Mangrove prop-roots, sandy patches, and sea-grass beds areas are not visited.

Once in the water, the fishers use flashlights to 'blind' lobsters and collect them. Daily fishing activities last between 6 to 7 h. Fishing activity peaks during the summer holidays (July and August, even if the national lobster fishery remains closed) and usually Wednesdays and Thursdays or Thursdays and Fridays, in preparation for weekend buyers (mainly tourists visiting the area) and to satisfy local restaurant demands.

Population characteristics

A sample of 163 spiny lobsters was examined. Of these, 93.3% (152) were *P. guttatus*, 6.1% (10) *P. argus*, and only 0.6% (1) *P. laevicauda*. Of the 152 *P. guttatus* measured, 59.2% (90) were males and 40.8% (62) were females. Thus, the sex ratio favoured males in a 1.45:1 ratio. The size of landed individuals ranged from 42.1 to 75.6 mm CL (Fig. 2), average 60.9 ± 6.9 mm CL. The mean size of females was 56.7 ± 5.6 mm CL; males averaged 63.9 ± 6.1 mm CL (Fig. 3). Of the females, 51.6% (32) were sexually mature (bore either eggs or an eroded spermatophore). Of the 62 females, 48.4% (30) were ovigerous and 30% (9) of these showed signs that the egg mass had been removed by the fisher. Egg mass removal by a fisher was identified by the presence of inconspicuous remains of egg patches. The smallest ovigerous female measured 48 mm CL; the

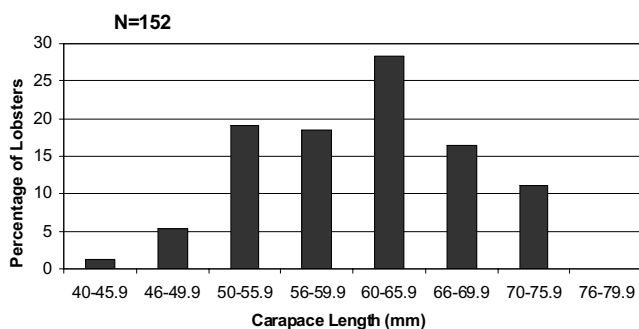


Fig. 2. Size distribution of the total sample of the lobster *Panulirus guttatus* captured by free divers from the Morrocoy National Park, Venezuela, during September to December 1999.

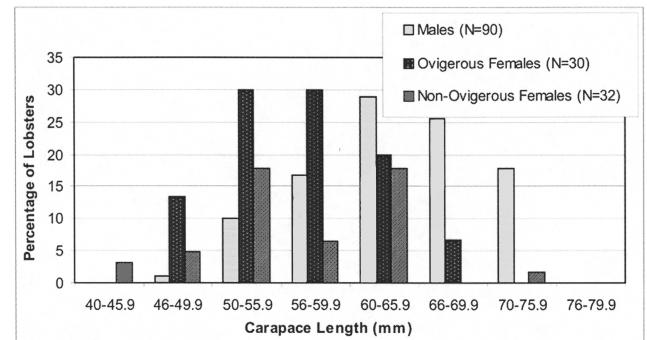


Fig. 3. Size distribution of female (ovigerous and nonovigerous) and male *Panulirus guttatus* from the Morrocoy National Park, Venezuela, captured during September to December 1999.

smallest female bearing an eroded spermatophore was 42.1 mm CL. The carapace length of the 10 recorded *P. argus* ranged between 55.0 and 88.0 mm. No *P. argus* were ovigerous, nor did they bear spermatophores. The only *P. laevicauda* found was a male that measured 70.0 mm CL.

Discussion

The Venezuelan lobster-management plan is based on the information collected for just one species (*P. argus*) at one site (Los Roques Archipelago National Park). Therefore, the fishery for *P. guttatus* is deemed illegal because even mature individuals are below the legal size limit, and that fishery is concentrated during the closed season for *P. argus*. The illegal nature of the *P. guttatus* fishery prevents the collection of valuable information about *P. guttatus* and the characteristics of the fishery.

As in the Morrocoy National Park, fishers in Bermuda, Florida, Honduras, and Mexico capture *P. guttatus* in shallow waters (no more than 6 m deep). They search for *P. guttatus* in the outer reef areas, mainly during the first three lunar quarters, which is its period of maximum activity (Caillouet *et al.* 1971; Briones-Fourzán 1991, 1995; Evans *et al.* 1995b; Sharp *et al.* 1997; Tewfik *et al.* 1998; Briones-Fourzán and Contreras-Ortiz 1999).

Of the lobsters landed during the present study period, 93.3% belonged to *P. guttatus*, which indicates that this is the current target species of the fishery. Unfortunately, there is no recorded history of the evolution of the spiny lobster fishery in the area, but the fishery for *P. guttatus* may have evolved because of a decline in the landings of *P. argus*. According to some fishers, *P. argus* used to be much more important and abundant in the park. Presently, *P. argus* is only found in deeper areas (>20 m), which are more difficult to reach by free diving. Even though only a few individuals of *P. argus* were measured, all were well below the minimum legal landing size of 120 mm CL. Either the fishermen are using their argument (lack of legislation for *P. guttatus*) to ignore the current management plan for

Table 1. Average lengths of females and males of the spotted spiny lobster, *Panulirus guttatus*, harvested from different sites in the Caribbean

Location	Carapace length (mm)		Capture method	Reference
	Female	Male		
Bermuda	70.0	79.0	Traps	Sutcliffe 1953
Bermuda	61.0	65.9	Traps	Evans <i>et al.</i> 1995b
Florida	51.9	58.4	Skin diving	Sharp <i>et al.</i> 1997
Martinique	51.3	58.7	Trammel nets	Marfin 1978
Mexico	68.2	73.2	Traps	Padilla-Ramos and Briones-Fourzán 1997
Mexico	54.8	61.1	Traps	Briones-Fourzán and Contreras-Ortiz 1999
Venezuela	56.7	63.9	Skin diving	Present study

P. argus, or they are not differentiating between the species, which is highly unlikely.

The mean size of *P. guttatus* landed in the Morrocoy National Park (60.9 mm CL) was smaller than that reported in Mexico (71.3 mm CL, Padilla-Ramos and Briones-Fourzán 1997) and Bermuda (79 mm CL, Sutcliffe 1953; 63 mm CL, Evans and Lockwood 1994) but larger than that recorded in Florida (53.6 mm CL; Sharp *et al.* 1997). The Florida study (Sharp *et al.* 1997) was a fishery-independent survey, which may explain the smaller mean size. Females tended to be smaller than males (average 56.7 and 63.9 mm CL, respectively). Similar findings have been observed in Bermuda (Sutcliffe 1953; Evans *et al.* 1996), Florida (Sharp *et al.* 1997), Martinique (Marfin 1978), and Mexico (Padilla-Ramos and Briones-Fourzán 1997; Briones-Fourzán and Contreras-Ortiz 1999) (Table 1).

As found in the present study, the sex ratio of *P. guttatus* has been reported to be skewed toward males (1.20:1, Sharp *et al.* 1997; 1.71:1, Padilla-Ramos and Briones-Fourzán 1997; 1.80:1 Briones-Fourzán and Contreras-Ortiz 1999). In some cases, the skew has been large (10:1, Sutcliffe 1953; 12:1, Evans and Lockwood 1994). Only Sharp *et al.* (1997) observed a female-biased sex ratio (1:3) during nocturnal captures. The disparity in the sex ratio of *P. guttatus* has been associated with the method of capture (traps vs. handpicking; Briones-Fourzán 1991), lower catchability (Evans and Lockwood 1994), differential male and female movements associated with their reproductive activity (Briones-Fourzán and Contreras-Ortiz 1999), and lower abundance due to differential mortality (Evans *et al.* 1995; Sharp *et al.* 1997). The evolutionary mechanics of skewed sex ratios in *P. guttatus*, if they exist, require further study.

The smallest ovigerous female found in the present study was 48 mm CL; the smallest reproductive female (bearing a spermatophore) measured 42.1 mm CL (smallest female found). These values were higher than those observed for ovigerous females in Florida (36 mm CL, Caillouet *et al.* 1971; 38 mm CL Sharp *et al.* 1997) and Mexico (36 mm CL, Briones-Fourzán 1991; 40 mm CL, Briones-Fourzán and

Contreras-Ortiz 1999) but lower than those recorded in Bermuda (55.5 mm CL, Evans *et al.* 1995b). The small number of egg-bearing females observed in the present study (30) does not allow us to provide a robust estimate of the size at which at least 50% of the females are egg-bearing.

Some studies suggest that *P. guttatus* spawns year-round (Florida, Caillouet *et al.* 1971; Sharp *et al.* 1997; Mexico, Padilla-Ramos and Briones-Fourzán 1997; Briones-Fourzán and Contreras-Ortiz 1999), whereas others indicate that reproduction is seasonal (Bermuda, Sutcliffe 1953; Evans and Lockwood 1994). Reproductively active females (with eggs or spermatophores) were found during each of the four months in the present study (September to December), and according to the local fishers the female spotted spiny lobsters are found berried year-round. Monthly surveys, however, should be conducted to determine both the extent and peak of reproductive activity.

A total of 30% of the ovigerous females showed signs of egg-mass removal. Law forbids landing egg-bearing females. It seems that fishers try to minimize the fault of harvesting undersized lobsters by removing the egg mass. A future management plan should include an educational program to correct such damaging fishing practices.

Management plans that include *P. guttatus* have existed in Martinique and Mexico since 1975 and 1987, respectively (Farrugio 1975; Briones-Fourzán 1991). The minimum landing size of 60 mm for *P. guttatus* was proposed to protect approximately 50% of the adult population. We suggest a similar strategy for Venezuela. In this way, the fisher will not be penalized with a minimum capture length that is excessive for this species and the fishery will be carried out less furtively, so valuable information will be easier to gather. Annual and seasonal studies are needed for the Venezuelan *P. guttatus* fishery to improve our knowledge of its biology, ecology, and economic significance. The information from this preliminary study should be useful in developing a fisheries management plan for *P. guttatus*.

Acknowledgments

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